

A Model-Free Explorer for ARC-AGI-3

Architecture, Ablations, and What the Metric Actually Rewards

TGAER — ARC Prize 2026, ARC-AGI-3 track

29 September 2026

Abstract

We report on a model-free exploration agent for ARC-AGI-3, a benchmark of interactive grid games whose rules are never stated. Over the period covered here the agent was measured against a 25-game local harness with a 0.0300 pp noise floor, and submitted to the Kaggle leaderboard, which scores *110 games the agent has never seen*. Our headline finding is negative and specific: of fifteen gated mechanism changes, one shipped with a measurable local gain, one shipped bit-identical and moved the public score, and thirteen were falsified. We decompose the score to show that *all* remaining local headroom is action efficiency on games already solved rather than new games solved, and that the two games holding 64% of that headroom fail for opposite reasons requiring opposite fixes. We also show that the local benchmark is a poor proxy in *both* directions, which reframes what a submission should be selected for.

1 The metric, and why it drives every decision

A level scores $\min(\text{cap}, (h/a)^2)$ where h is the human baseline action count and a is ours. A game averages its levels *weighted by level index over all of its levels*, so an uncleared level contributes zero to the numerator while its weight remains in the denominator. Three consequences set the whole research agenda:

1. **Quadratic in efficiency.** Being $10\times$ slower than a human costs $100\times$ the score. Level *count* is close to irrelevant by comparison.
2. **Later levels are worth more.** In an 8-level game level 1 is $1/36$ of the score and levels 2–8 are $35/36$. Exploration early is cheap; exploitation late is valuable.
3. **Actions on a level never cleared are free.** The numerator is zero either way, so truncation costs nothing and monotonicity in the action budget holds.

2 Evaluation protocol

Local measurements use 25 public games at 600 actions, five seeds per arm, reporting Relative Human Action Efficiency (RHAЕ). The gate requires three things, and a candidate failing any one is rejected: (i) no game may score less often and no game may lose a level, (ii) wall clock within $1.25\times$ of baseline with both arms projecting inside the kernel’s deadline, and (iii) an argument for why the change should help *off-roster*. Baseline is $0.3520\% \pm 0.0280$, per-seed $[0.3827, 0.3827, 0.3316, 0.3316, 0.3316]$; the noise floor is 0.0300 pp.

Two protocol lessons were learned the hard way and are recorded because they changed conclusions. First, **a candidate standard deviation of exactly zero is a defect report, not a clean result**: the entire seed variance of the 25-game suite comes from one game (1p85) reaching level 3 on two seeds, so $\sigma \rightarrow 0$ means that outcome was pinned. Second, **the headline verdict is not sufficient**: one candidate read “inside noise” while 1p85 lost 0.8 levels on average, just under the level-drop threshold. The per-game difference must be read every time.

3 Where the headroom is

Each game’s **cap** is the score it would post at *perfect* efficiency on the levels it currently clears. Averaged over the roster this is the ceiling reachable by pure speed, with no additional level solved. At 6000 actions over three seeds:

game	levels	env	% cap	% capture
lp85	3/8	3.6330	19.444	18.7%
tu93	3/9	0.0455	13.333	0.3%
m0r0	1/6	0.0024	4.762	0.1%
sp80	1/6	0.0345	4.762	0.7%
g50t	1/7	3.5714	3.571	100.0%
ar25	1/8	1.8701	2.778	67.3%
s5i5	1/8	0.0006	2.778	0.0%
total				0.3663 2.0571 17.8%

18 of 25 games score zero, and their cap is zero too, so coverage work returns nothing until level 1 actually falls. Everything reachable is efficiency on games already won. **g50t** is already at human pace and **ar25** at 67%, so the two games supplying most of today’s score have nothing left to give.

4 Diagnosis: stranded versus flooded

Instrumenting the agent’s own state graph shows the two games holding 64% of the ceiling fail in opposite ways.

tu93 L2 lp85 L4

steps on the level	5424 / 6000	4750 / 6000
no route to a frontier	3581 (66%)	0 (0%)
current state is itself a frontier	474	3764 (79%)
states discovered	frozen at 159	growing to 1647
untested pairs remaining	frozen at 6	growing to 15950

tu93 is stranded: from step 2500 to 6000 it discovers no new state and drains no untested pair, with six pairs sitting at three states it cannot reach. The adjacency is directed and holds only observed transitions, so the agent learns $a \xrightarrow{X} b$ but never the way back unless it happens to take the inverse at b ; inside a loop it never does. **lp85 is flooded**: it never fails to find a route because nearly every state is itself a frontier, and it generates states faster than it can drain them. A reachability sweep confirms all seven scoring games are budget-limited rather than out of reachable states.

5 Ablation ledger

Every row was gated at five seeds against the noise floor. Rejections are the majority and are the substance of the work.

mechanism	Δ RHAE	outcome
action budget 6000 $\rightarrow$ 12000	+0.0001 pp	shipped; monotone, so free
colour-agnostic role inference	+0.0000 pp	shipped bit-identical ; moved public +0.01
probe cap 4 $\rightarrow$ 1	+0.1959 pp	shipped; largest local gain recorded
defer irreversible edges	−0.1659 pp	rejected, 8.4σ ; three games stop scoring
drop the field-box crop	−0.1661 pp	rejected; crop is the denoiser
re-validate the avatar latch	—	rejected; tu93 2 $\rightarrow$ 1 levels
(colour, size) compression	—	not built; candidates collapse only $1.4\times$
class-generalised inert demotion	+0.0000 pp	rejected; bit-identical on all seeds
effect (inertness) prediction	+0.0065 pp	closed by ceiling : $5\times$ under the floor
per-level scoping of the stuck test	−0.0012 pp	rejected; tu93 5/5 seeds $\rightarrow$ 0/5
fresh novelty window on respawn	−0.0012 pp	rejected; <i>identical</i> to the row above
verify a routed step’s destination	+0.0000 pp	bit-identical; routes never mislead
learned inverse edges	−0.0014 pp	rejected; only tu93 moved, and it died
carry inert evidence across a level	−0.0204 pp	rejected; $\sigma = 0$, lp85 pinned
same, capped at 1	−0.0204 pp	rejected; <i>bit-identical</i> to uncapped
same, controls only	+0.0000 pp	no-op; 0 of 25 games moved
novelty judged against the game’s own ceiling	−0.0002 pp	submitted ; passes all three gates

5.1 Three results worth more than the fixes would have been

Effect prediction is closed by a bound, not an experiment. A *perfect oracle* for “this action changes nothing” is worth +0.0065 pp against a 0.0300 pp floor — $5\times$ below measurable. The value of predicting that an action does nothing is bounded by the cost of finding out, which is one action. Three games have *zero* inert actions in 600 steps. This does not close mechanisms that avoid *expensive* mistakes.

tu93 is a knife edge, so its headroom is not capturable. Three independent routing mechanisms each took it from scoring in 5/5 seeds to 0/5, and the cleanest of them moved *no other game at all*. Its wins depend on an accident: a death lowers the completed-level count, which reopens a gate while the novelty window still holds pre-death evidence, so an escape switch fires on the respawned board. Its nominal 25.9% of the ceiling cannot be reached by changing how it moves, because reaching its stranded states costs the levels it already has.

Nothing in the inert map transfers across a level. Its informative half is clicks, whose keys embed (row, col) and therefore name a board; its board-free half is 4–6 controls relearned within a handful of steps. Hence the membership rule: knowledge belongs in a cross-level tier only if its key is board-free *and* re-earning it is expensive.

6 Constant sensitivity

A real mechanism improves over a *range*; a lucky rollout spikes at one value with baseline either side. Sweeping the module-level tuned constants:

constant response reading

PROBE_LIMIT	0.3825 → 0.3078 → 0.3073 → 0.1865	monotone	to a floor at 1, not a spike
CHURN_FRACTION	0.1861, 0.3800, 0.3825 , 0.3314, 0.3314		plateau over 0.4–0.5

The sweep tool labelled PROBE_LIMIT a spike because only one value beats baseline, but its own definition of a spike requires baseline *on both sides*. The response is monotone and 1 sits at the boundary; the code further documents that 0 and 1 are identical on all 25 games, with 1 retained to keep the mechanism alive for an unseen game. The constant is therefore at a natural floor rather than fitted to noise.

7 Prior art, filtered by the rules we actually play under

Internet access is disabled in the scored kernel, which are the competition’s own terms. Every published result above ours that uses a frontier API is therefore a research result on the public set, not a submittable design.

system score model in-kernel?

Tycho	100.0	RHAE, all 183 levels	GPT-5.6 / Opus 5	no
Executable World Models	58.12%	RHAE	GPT-5.5	no
Duck harness (winner)	1.21%	Qwen 3.6 27B FP8		yes
AERA	0.30 (private, 55 games)	Qwen2.5- 0.5B		yes
ours	0.13–0.14	none (model-free)		yes

The benchmark is close to solved by frontier coding agents; the *competition* is not, because it forbids them. The sharpest datum is not the winner’s 9× margin but that **AERA beats us roughly 2× using a 0.5B model**, which retires compute and model size as explanations and leaves architecture. Both kernel-feasible systems above us are LLM-as-programmer designs. An ablation of the world-model family reports that *verification* ranks first among its mechanisms and, critically, succeeds at *lower model effort* — the regime a local quantised model must work in — while the executable world model is “not universally beneficial”.

8 The local benchmark is a poor proxy in both directions

submission change local RHAE public

v76	probe cap 4 → 1	0.1561 → 0.3520% (2.25×)	0.13
v77	12000 actions + colour-agnostic roles	unchanged, bit-identical	0.14
	repeat	<i>byte-identical rebuild of v77</i>	— 0.14
v78	novelty against the game’s own ceiling	−0.0002 pp	pending

A $2.25\times$ local gain moved the public score by zero; a local *no-op* moved it by +0.01. The identical rebuild scoring identically is the first evidence that the public score is reproducible at 0.01 resolution, which makes v77’s +0.01 real. The caveat is that this rests on a single pair, so equality bounds the variance weakly.

The operative conclusion is that a flat local gate is *not* a reason to withhold a submission, and a local gain is not a reason to make one. What earns a submission is a named mechanism by which the change differs off-roster. v77 removed a dependency on public-set colour indices; v78 removes a threshold fitted inside a 0.11–0.18 gap measured on five of the 25 games that are never scored.

9 Conclusions

1. All remaining local headroom is **action efficiency on games already solved**. Coverage work returns nothing while 18 games score zero.
2. The two games holding that headroom need **opposite** fixes, which explains why single mechanisms aimed at “exploration” kept failing.
3. **tu93**’s share of the ceiling is **not capturable** by routing changes, and it has decided three gates on its own. A candidate should be required to move a game other than **tu93**.
4. **1p85** is an **ordering** problem, not a search problem: 15950 untested pairs where which candidate is tried first is the whole game.
5. Ordering under a budget is what a value model or a program-writing agent supplies, and is the one architecture we have never actually tested. Our earlier 27B failure tested LLM-as-*policy*, the weak variant.

Next steps, in order. Confirm the kernel’s runtime limit, which is not stated in the official documentation while our code assumes 7.5 h. Extend oracle labels past level 0 so the transfer metric (actions-per-level regressed on level index) becomes decisive. Then resume the REPL branches toward the published winner’s design, sized to the confirmed accelerator.